

# Nara Ávila

## Master's Student

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|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| Education          | <b>M.Sc. in Nuclear Physics</b> <a href="#">[link]</a><br>Physics Institute (IF)<br>University of São Paulo (USP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2025 – Present                                  |
|                    | <b>B.Sc. in Computer Engineering</b> <a href="#">[link]</a><br>São Carlos School of Engineering (EESC)<br>University of São Paulo (USP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2006 – 2011                                     |
| Academic Positions | <b>Research Collaborator</b> <a href="#">[link]</a><br>40 hours / week<br>Project: Quantum Computing Applied to Molecular Modeling<br>Department of Materials Physics and Mechanics (DFMT)<br>Institute of Physics (IF), University of São Paulo (USP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2023 – 2025                                     |
|                    | <b>Teaching Assistant</b> <a href="#">[link]</a><br>8 hours / week<br>• SME0300 Numerical Methods I<br>Institute of Mathematics and Computer Sciences (ICMC)<br>University of São Paulo (USP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Fall 2009                                       |
|                    | <b>Teaching Assistant</b> <a href="#">[link]</a><br>8 hours / week<br>• SME0601 Numerical Methods II<br>Institute of Mathematics and Computer Sciences (ICMC)<br>University of São Paulo (USP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Spring 2009                                     |
|                    | <b>Awards</b> <ul style="list-style-type: none"><li>• Best Presentation in Environmental Management System (SHS0436)<ul style="list-style-type: none"><li>– Recognized for an impactful final course presentation entitled “Harmonic and Disharmonic Systems,” due to its combination of visual simplicity and strong conceptual content. Using images alone, the presentation explored in detail the diversity, complexity, and beauty of harmonic systems in nature—including coral ecosystems and harmonious ecological relationships such as mutualism, commensalism, and epiphytism—in contrast with human systems that disrupt environmental balance, illustrated through examples of waste accumulation, pollution, environmental degradation, and pollution-related wildlife mortality, alongside other disharmonic relationships also found in nature, such as parasitism and disease. Received the maximum final grade (10.0/10.0).</li></ul></li><li>• 9th place in the ACM-ICPC / SBC Brazilian Programming Contest <a href="#">[link]</a><ul style="list-style-type: none"><li>– Competed as a member of team “Greve de Acertos” in the São Paulo State round of the Brazilian programming competition organized by the Brazilian Computer Society (SBC) as part of the International Collegiate Programming Contest (ICPC).</li></ul></li><li>• 6th place in the IPhO / SBF Brazilian Physics Olympiad <a href="#">[link]</a><ul style="list-style-type: none"><li>– National physics olympiad organized by the Brazilian Physical Society (SBF) and part of Brazil’s selection process for the International Physics Olympiad (IPhO).</li></ul></li><li>• 18th place in the OBM / XI OMEG <a href="#">[link]</a><ul style="list-style-type: none"><li>– Mathematics olympiad competition associated with the Brazilian Mathematical Olympiad (OBM), Brazil’s national mathematics olympiad and part of the country’s selection pathway for the International Mathematical Olympiad (IMO).</li></ul></li></ul> | 2009<br>9th – 2008<br>6th – 2003<br>18th – 2002 |

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| Grants                | <b>Entrepreneurship Workshop (PRPG/USP)</b> <a href="#">[link]</a> 2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                       | Project Title: StrategiQ — AI-Based SaaS for Quantum Computing Governance Consulting<br>Role: Founder and Project Lead of a three-member team conducting market research and developing the business plan for StrategiQ developed through the Scientist Entrepreneur Training (DPG5011)<br>Funding Agency: Office of the Provost for Graduate Studies, University of São Paulo (PRPG/USP)<br>Total Amount: R\$3,720.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                       | <b>132734/2025-7</b> <a href="#">[link]</a> 2025–2027<br>Program: Master’s Scholarship (GM)<br>Funding Agency: National Council for Scientific and Technological Development (CNPq)<br>Host Institution: University of São Paulo (USP)<br>Total Amount: R\$50,400.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                       | <b>409642/2022-3</b> <a href="#">[link]</a> 2023–2024<br>Project Title: Quantum Computing Applied to Molecular Modeling<br>Program: Technological Development in ICTs (DTC-F)<br>Funding Agency: National Council for Scientific and Technological Development (CNPq)<br>Scholarship Process: 301681/2023-6<br>Total Amount: R\$36,300.00                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Software              | <ol style="list-style-type: none"> <li>MORAES, N. A. <i>Molecular Symmetry</i>. Web application for parsing molecular geometries from XYZ files and generating molecular symmetry reports. <a href="#">[link]</a>.</li> <li>MORAES, N. A. <i>Quantum Battleship</i>. Quantum computing game that allows users to play Battleship using real quantum computers. <a href="#">[link]</a>.</li> <li>MORAES, N. A. <i>GitHub Release and Milestone CLI</i>. Command-line tool for automating GitHub releases, milestones, and software development cycles. <a href="#">[link]</a>.</li> <li>MORAES, N. A. <i>Zotero Mirroring Plugin</i>. Zotero plugin for mirroring the hierarchical structure of a Zotero library onto the local filesystem. <a href="#">[link]</a>.</li> <li>MORAES, N. A.; LUZUM, M. W. <i>hic-bayes</i>. Python library for Bayesian parameter estimation and Gaussian-process emulation in relativistic heavy-ion collision studies. Implements PCA-based dimensionality reduction, Gaussian-process emulators, multivariate likelihoods, MCMC sampling, posterior analysis, and multi-fidelity workflows. <a href="#">[link]</a>.</li> </ol> |
| Leadership Experience | <b>Founder &amp; Team Lead</b> 2025<br>DPG5011 – Entrepreneurship, University of São Paulo (USP) <ul style="list-style-type: none"> <li>• Founded and led a six-member interdisciplinary team in the development of an innovative startup focused on quantum computing.</li> <li>• Led the team through the development of the startup concept, including problem definition, business strategy, and the application of quantum technologies to industry-oriented solutions.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                       | <b>Undergraduate Research Mentor</b> 2025<br>Institute of Physics, University of São Paulo (IFUSP) <ul style="list-style-type: none"> <li>• Mentored a group of five undergraduate students in scientific initiation projects in quantum computing, providing guidance throughout their research activities.</li> <li>• Supervised the development of individual research projects presented by the students at the Undergraduate Research Symposium.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Professional Diplomas

Postgraduate Specialization in Financial and Capital Markets [\[link\]](#) 2021 – 2022  
• Quantitative methods, corporate finance, financing, and financial products and markets  
*Mackenzie Presbyterian University*

## Professional Experience

Software Engineering Specialist 2020 – 2022  
Open Co

- Developed and optimized backend services for a digital banking platform, including the first installment payment solution via PIX.
- Proposed and prototyped the migration of a credit-analysis workload from Kubernetes (EKS) to AWS Lambda, reducing response time and operational costs by approximately 90%.

Senior Software Engineer 2019 – 2021  
TIDEXA

- Contributed to the migration of a payroll system from Oracle Forms/PL–SQL to a Java-based platform.
- Led technical decisions and development activities within the engineering team.

Competitive Public-Sector Examination Preparation 2016 – 2019  
Independent, full-time study

- Multidisciplinary preparation for competitive Brazilian public-sector examinations, including Law, Accounting, Economics, Public Finance, Public Administration, and Business Administration.

Mid-Level Software Engineer XXXX – 2016  
Itaotec – Grupo Itaotec

Junior Software Engineer 2012 – XXXX  
Itaotec – Grupo Itaotec

- Redesigned a business-critical software application using software design patterns and engineering best practices, improving usability, reliability, and maintainability.
- The application was deployed across more than 5,000 branches of a major Brazilian bank.

## Talks & Presentations

### Conference Presentations

1. “The Effect of Correlated Experimental Uncertainties on Bayesian Inference of Quark–Gluon Plasma Properties.” Poster presentation at the PHENOEXP Workshop, August 2026. [\[link\]](#).
2. “Study of the Effect of Correlated Experimental Uncertainty on Bayesian Inference of Quark–Gluon Plasma Properties.” Oral presentation at the XXXIII Reunião de Trabalho sobre Interações Hadrônicas (RETINHA XXXIII), June 2026. [\[link\]](#).
3. “Risk Management and IT Governance in the Adoption of Quantum Computing: A COBIT 5-Based Conceptual Model.” Oral presentation at the 5th Congress on Risk Management and ERM, FEA–USP, December 2025. [\[link\]](#).
4. “Exploring the Potential of Quantum Computing in Materials Science.” Oral presentation (co-author), Autumn Meeting of the Brazilian Physical Society, 2025. [\[link\]](#).

### Industry & Professional Talks

1. “Perspectives on Quantum Computing.” Talk at Rio Innovation Week – AI and Meta World, 2024. [\[link\]](#).
2. “Quantum Search Algorithms.” Talk at Open Co Tech Day, 2022. [\[link\]](#).

3. “Shift Your Programming Paradigm to a Quantum Mindset.” Talk at The Developer’s Conference (TDC Business), 2022.  
[\[link\]](#).
4. “Fundamentals of Quantum Computing.” Talk at Open Co Tech Day, 2021.  
[\[link\]](#).
5. “Technical Software Documentation.” Seminar at ITAUTEC, 2014.  
[\[link\]](#).

### **Selected Highlights**

School Environmental Research Project – Municipal Waste Management 1999  
Goiânia, Goiás, Brazil

- Following a field visit to the Goiânia Municipal Landfill, interviewed Pedro Wilson Guimarães, then Federal Deputy for the State of Goiás, for a school research project on municipal solid waste management.
- Discussed waste-management challenges and technologies, including landfill biogas recovery and international approaches to waste incineration.
- Wilson later served as Mayor of Goiânia (2001–2004), where his administration established municipal selective waste collection through legislation enacted in 2002.

### **Extracurricular Activities**

1st Place – Brazil Interstate Taekwon-Do ITF Competition [\[link\]](#) August 2023

- Two Gold medals in Sparring and Patterns (Tul).
- Buzios. Rio de Janeiro, Brazil.

1st Place – Children’s Drawing Contest [\[art portfolio\]](#) October 2000  
Claro – Telecommunications Company, Brazil

- First-place award in a Children’s Day drawing competition organized by Claro for employees’ children.